

Viraj Induruwa

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ABOUT MYSELF

Full-Stack Software Engineer with strong expertise in backend development using Python, FastAPI, and Django, focused on building scalable APIs, automation systems, and enterprise applications. Experienced in frontend development and integrating AI and machine learning capabilities into real-world projects, including conversational AI, computer vision, and predictive analytics.

SKILLS

- **Technical Expertise:** Python, JavaScript, FastAPI, Flask, Django, React, TensorFlow, PyTorch, Keras, Scikit-Learn, OpenCV, YOLO, Hugging Face Transformers, PostgreSQL, MySQL, Oracle DB, Docker, AWS, Linux, Azure Active Directory, REST APIs, JWT, OAuth, OCR, RAG Systems
- **AI / Machine Learning:** Deep Learning, Computer Vision, NLP, CNNs, Predictive Analytics, LLM Integration, Emotion Detection, Stress Monitoring AI, Multi-modal AI Systems, Model Deployment, ML Inference APIs
- **Cloud, Tools & Platforms:** Kafka, Git, GitHub, Bitbucket, Postman, Playwright, Streamlit, Tailwind CSS, AWS EC2, AWS Amplify, AWS RDS, AWS S3 Bucket, AWS Lambda, Azure AD, Linux Servers
- **Professional Skills:** Problem Solving, Analytical Thinking, Team Collaboration, Communication, Leadership, Adaptability, Time Management, Project Management, Attention to Detail

WORK EXPERIENCE

Freelance Software Engineer – Sri Lanka Telecom (SLT) Oct 2025 – Present

- Continued development and enhancement of the **Sales Incentive Automation System** on SLT's internal infrastructure
- Modules developed during internship and freelance period were successfully deployed on **Linux servers** within the corporate intranet environment, integrating **Azure AD authentication** and **Oracle DB**
- Led the development of a new **Manager Incentive Module**, automating managerial-level incentive calculations with advanced business logic
- Transitioned business workflows from manual Excel and Tableau-based processing to a centralized automated system
- Enhanced operational efficiency by **80%** while significantly reducing human calculation errors and improving reporting turnaround time

Intern Software Engineer – Sri Lanka Telecom (SLT) Apr 2025 – Oct 2025

- Contributed to the development of the **Sales Incentive Automation System** to replace manual Excel-based workflows used by sales and dealer management teams
- Designed and implemented a **Python-based console application** to automate complex sales incentive and dealer commission calculations
- Developed backend APIs using **Django and FastAPI**, integrating with **PostgreSQL and Oracle databases**
- Implemented secure authentication and role-based access control using **Azure Active Directory (Azure AD), Google OAuth, and JWT**
- Improved calculation accuracy and reduced manual processing time by approximately **90%**

PROJECTS

Mindful-AI - Stress Detection & Monitor Chatbot

- Developed an AI-powered conversational stress detection and monitoring platform using FastAPI and React
- Built a multi-turn AI chatbot with emotion-aware conversations and stress severity classification using PyTorch, HuggingFace Transformers, and Scikit-Learn
- Tools Used: React, FastAPI, PyTorch, HuggingFace Transformers, Scikit-Learn, PostgreSQL, Tailwind CSS

Gemstone ML Unified API: Identification & Cut Advisor

- Developed a unified AI-powered gemstone identification and cut recommendation API for the gemological industry
- Built a multi-modal prediction system combining CNN image analysis with physical gemstone properties for gemstone type, authenticity, and origin detection
- Designed an intelligent cut advisor using CatBoost and XGBoost to recommend optimal gemstone cuts and precision cutting parameters
- Tools Used: FastAPI, TensorFlow, Keras, CatBoost, XGBoost, Scikit-Learn, Pandas

AI-Powered Hospital Management System

- Developed an AI-powered Hospital Management System integrating predictive analytics for inventory, equipment maintenance, and patient load forecasting
- Built four machine learning engines for drug expiry prediction, inventory consumption forecasting, equipment maintenance analysis, and patient traffic forecasting
- Tools Used: React, FastAPI, PostgreSQL, CatBoost, Scikit-Learn, AWS EC2, Tailwind CSS

Cancer Detective Web App

- Built Django with React web app for early detection of skin, lung, and leukemia cancers using 30K+ medical images
- Achieved 93% (skin), 98% (lung), and 88% (leukemia) detection accuracy with deep learning models
- Developed and containerized FastAPI-based ML inference APIs using Docker, then deployed them on AWS Lambda for scalable real-time cancer prediction services
- Tools Used: Django, React, TensorFlow, Keras

EDUCATION

Horizon Campus

BSc (Hons) in Information Technology

Malabe, Sri Lanka

May 2021 – Oct 2025

CGPA: 3.30/4.00 | SECOND UPPER CLASS HONOURS

Relevant Coursework: Software Engineering, Operating Systems, Artificial Intelligence, Data Science

Website: horizoncampus.edu.lk 

LANGUAGE SKILLS

Mother Tongue(s): Sinhala

Other Language(s): English - Professional Working Proficiency

I am Passionate about building scalable backend systems and full-stack applications with modern technologies. Continuously expanding expertise in Python backend development, AI integration, and cloud-based solutions while contributing to impactful and innovative software projects.